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Evaluating Super-Resolution Techniques and Metrics for Enhancing Nightlight Satellite Imagery

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1 Introduction

Artificial lighting observed from space provides important information on human activity across the globe. Instruments such as the *Visible Infrared Imaging Radiometer Suite* [16] capture the radiance emitted by cities, infrastructure, and other illumination sources, allowing researchers to analyze urbanization patterns, economic activity, and disruptions caused by disasters. Widely used NTL products, including NASA’s Black Marble Nighttime Lights, offer consistent global coverage with high temporal frequency. However, their spatial resolution remains relatively coarse compared to many other Earth observation datasets, limiting the ability to analyze fine-scale urban structures and smaller settlements. Super-resolution (SR) techniques offer a potential approach to overcome this limitation by reconstructing higher-resolution imagery from lower-resolution observations. While many SR methods have been proposed in the computer vision literature, their evaluation typically relies on generic image quality metrics whose relevance for NTL data remains unclear. This project investigates the application of super-resolution methods to nighttime light imagery and evaluates different quantitative metrics for assessing SR performance in this remote sensing context.

2 Related Work

2.1 Nighttime Light Satellite Imagery and Its Applications

Satellite nighttime light (NTL) imagery has become an important data source for monitoring human activity and infrastructure on a global scale. Nighttime light observed from space provides information on economic activity, urbanization, electrification, and population distribution. A comprehensive overview of the applications and limitations of NTL data is provided by *Remote Sensing of Night Lights: A Review and an Outlook for the Future* [1], which highlights the wide range of domains in which nightlight imagery has been used, including urban monitoring, economic analysis, and disaster response.

One of the most widely used satellite NTL datasets is the *NASA Black Marble Nighttime Lights Dataset* [19]. It provides nightly global observations of nighttime radiance and includes atmospheric correction, cloud masking, and radiometric calibration. It has the advantage of high temporal resolution, which enables consistent and long-term monitoring. It is used for example in *Monitoring Disaster related power outages using nasa black marble nighttime light product* [2]. Another widely used data set is derived from the Luojia satellite, which has a high spatial resolution but a poor temporal resolution. Luojia nighttime light imagery has been used to map urban extent, demonstrating that roads and intra-city lighting patterns that are not visible in lower-resolution sensors such as Suomi NPP VIIRS can be resolved [3].

Several studies have demonstrated the usefulness of NTL imagery for detecting disruptions in electrical infrastructure during natural disasters and humanitarian crises. For example, *Using Satellite Night Lights to Monitor Power Outages During Disasters* [2] shows that decreases in nighttime illumination can be used to identify power outages and track recovery following events such as hurricanes and earthquakes.

Despite these advances, many widely used NTL datasets such as Black Marble still have

relatively coarse spatial resolution. Super resolution is currently being explored as a solution to improving resolution for satellite imagery. In remote sensing, SR has been widely studied as a way to overcome the spatial resolution limitations of satellite sensors.

2.2 Super-Resolution in Remote Sensing

Super-resolution (SR) techniques aim to reconstruct high-resolution (HR) images from low-resolution (LR) observations. Surveys such as *Advancing Image Super-resolution Techniques in Remote Sensing: A Comprehensive Survey* [4] provide an overview of both traditional and deep learning-based SR methods for Earth observation imagery.

Early SR approaches relied primarily on interpolation techniques such as bicubic interpolation. While computationally efficient, these methods typically produce overly smooth images and fail to reconstruct fine spatial details. More recent work has focused on deep learning approaches that learn the mapping between LR and HR images using convolutional neural networks (CNNs).

Several remote sensing studies have adopted convolutional neural network architectures derived from the Efficient Sub-Pixel Convolutional Neural Network (ESPCN) introduced by Wenzhe Shi et al. (2016) [5], which uses sub-pixel convolution layers to efficiently reconstruct high-resolution images from low-resolution inputs. Variants of ESPCN have been applied to satellite imagery tasks such as building mapping and remote sensing image super-resolution, demonstrating strong performance while maintaining relatively lightweight architectures (e.g., Wang et al., 2019) [6]. More complex models have also been proposed, including generative adversarial network (GAN) based approaches such as *ESRGAN: Enhanced Super-Resolution Generative Adversarial Networks* [7]. These models can generate highly detailed textures but may introduce hallucinated features that do not correspond to the true underlying signal [7], which can be problematic for scientific remote sensing applications where radiometric accuracy is important.

Many super-resolution studies generate synthetic low-resolution images by degrading high-resolution images through operations such as Gaussian blurring and downsampling, allowing the creation of paired training data for supervised learning [8, 5]. Some works instead rely on corresponding observations from different satellites that provide imagery at different spatial resolutions. For example, the study *A comparative study of deep learning methods for super-resolution of NPP-VIIRS nighttime light images* uses paired observations from the Luojia-1 satellite and VIIRS Black Marble nighttime light data to train super-resolution models that map coarse-resolution VIIRS imagery to higher-resolution Luojia observations [9].

2.3 Evaluation Metrics for Super-Resolution

The evaluation of super-resolution algorithms typically relies on quantitative image quality metrics that compare reconstructed HR images with ground truth references. A comprehensive overview of these metrics is provided in *A Comprehensive Review of Image Super-Resolution Metrics* [10].

Among the most commonly used metrics are Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM). PSNR measures pixel-wise reconstruction fidelity based on the mean squared error between reconstructed and ground truth images. While widely used in

SR benchmarks, PSNR does not account for perceptual aspects of image quality and may not correlate well with visual similarity.

To address this limitation, perceptually motivated metrics such as the Structural Similarity Index (SSIM) were introduced in *Image Quality Assessment: From Error Visibility to Structural Similarity* [11]. SSIM evaluates similarity by comparing luminance, contrast, and structural information between images and has become a standard metric for image quality assessment in super-resolution research.

2.4 Evaluation of Super-Resolution for Remote Sensing

Evaluating super-resolution performance for remote sensing imagery presents unique challenges compared to natural image datasets. Remote sensing images often contain complex spatial structures, sensor-specific noise characteristics, and radiometric constraints that must be preserved during reconstruction.

Several studies have investigated SR evaluation specifically for Earth observation data. For example, *Remote Sensing Image Super-Resolution and Object Detection: Benchmark and State of the Art* [12] presents a benchmark framework for evaluating SR methods in the context of downstream tasks such as object detection. These studies demonstrate that improvements in traditional image quality metrics do not always translate into better performance for remote sensing applications.

Similarly, *Evaluating Super-Resolution Reconstruction of Satellite Images* [13] analyzes the effectiveness of different evaluation metrics for daylight satellite image reconstruction and highlights the importance of considering both spatial fidelity and radiometric accuracy when assessing SR performance.

Given these challenges, selecting appropriate evaluation metrics is particularly important for remote sensing super-resolution tasks. In the context of nighttime light imagery, where pixel intensities represent physical radiance measurements, the choice of evaluation metric may significantly influence the interpretation of reconstruction quality.

3 Dataset Creation

3.1 Data source

Nighttime light (NTL) imagery was obtained from the VIIRS VNP46A2 Nighttime Lights dataset. It provides nightly global observations of nighttime radiance at a resolution of 500 m that is moonlight and atmosphere corrected [14]. To ensure data quality, only observations with `Mandatory_Quality_flag = 0` were retained, which correspond to the highest-quality retrievals. It removes observations affected by clouds, stray light, or other atmospheric disturbances. After filtering, images within a one-month period were aggregated using a median composite to reduce remaining noise and sparse data. From this, the `Gap_Filled_DNB_BRDF_Corrected_NTL` band was taken, which represents calibrated nighttime radiance values.

Google Earth Engine [17] was used to collect the 128x128 pixels size images. To get a diverse dataset and ensure that the images contained enough information i.e light, locations where randomly sampled through binning. Locations contained in the World Cities Database [18]

were binned by population size. 2000 locations were uniformly sampled from these bins. Using the latitude and longitude information on these locations, the corresponding NTL images were retrieved from GEE. Before retrieval a check was done to ensure that there was no overlap between the tiles.

The 2000 sample dataset was split into a 80/10/10 training/validation/test split. The training set is used to train the learning-based SR model, while the validation set is used for monitoring training performance and computing evaluation metrics. The test set is reserved for final evaluation and comparison of super-resolution outputs.

4 Methods

The objective of this work is to evaluate different image quality metrics for nighttime light super-resolution. To do so, multiple SR outputs are generated from synthetic low-resolution inputs and compared against the original high-resolution images.

4.1 Synthetic Low-Resolution Generation

To create low-high resolution pairs, low-resolution images (LR) are synthetically generated from the high-resolution images (HR). Each HR image is first blurred using a Gaussian filter, then downsamples by a factor of two. The Gaussian blur approximates the point spread function of a sensor, while downsampling reduces spatial resolution. The resulting LR image is the input to the SR methods, while the original HR acts as the ground truth.

A compact mathematical form of this degradation process is:

$$I_{LR} = \downarrow_2 (G_\sigma * I_{HR}), \quad (1)$$

where I_{HR} is the original high-resolution image, G_σ is a Gaussian kernel with standard deviation σ , $*$ denotes convolution, and $\downarrow_2$ denotes downsampling by a factor of 2.

4.2 Super-Resolution Methods

A lightweight convolutional neural network based on the Efficient Sub-Pixel Convolutional Neural Network [15] architecture is implemented. Since the primary focus of this work is the evaluation of super-resolution metrics rather than the development of new SR algorithms, a lightweight and well-established architecture was chosen. The network consists of a small number of convolutional layers followed by pixel shuffle upsampling, which reconstructs the high-resolution image from learned feature representations. The model operates with a scale factor of 2, matching the degradation applied during LR generation.

5 Evaluation Metrics

The main goal of this study is to analyze how different metrics evaluate super-resolved nighttime light (NTL) imagery. Image super-resolution is typically assessed using image quality metrics that compare reconstructed images to high-resolution (HR) ground truth data. In the computer

vision literature, metrics such as PSNR and SSIM are widely used to evaluate reconstruction accuracy and perceptual similarity [21, 22]. However, NTL imagery differs from natural images because pixel values correspond to physically meaningful measurements of radiance rather than purely visual intensity patterns [23].

In related remote sensing and geophysical applications, including climate data super-resolution, evaluation often emphasizes radiance or value-based error metrics such as RMSE and bias, which quantify how accurately the reconstructed data preserves the magnitude of physical variables [24]. For NTL imagery, both spatial structure and radiometric fidelity are important: super-resolution should reconstruct realistic spatial lighting patterns while also preserving the magnitude and distribution of radiance values.

To capture these different aspects of reconstruction quality, this study evaluates a set of complementary metrics. PSNR and SSIM assess numerical and structural similarity between images, while MAE, RMSE, and bias measure radiometric reconstruction error. Correlation-based metrics further evaluate how well the spatial distribution of radiance is preserved. In addition to quantitative evaluation, qualitative visual comparisons between SR outputs and HR ground truth images are used to assess how well these metrics reflect visually meaningful differences in reconstruction quality.

5.1 PSNR

Peak Signal-to-Noise Ratio (PSNR) measures pixel-wise reconstruction accuracy based on the mean squared error between images and is one of the most commonly used metrics in super-resolution research.

$$\text{PSNR} = 10 \log_{10} \left(\frac{L^2}{\text{MSE}} \right), \quad (2)$$

where

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (x_i - y_i)^2, \quad (3)$$

L is the maximum possible pixel value, x_i is the HR pixel value, y_i is the SR pixel value, and N is the total number of pixels.

PSNR primarily captures overall numerical similarity between reconstructed and reference images. In the context of NTL imagery, it reflects how closely reconstructed radiance values match the ground truth at the pixel level. Because it is based on squared error, PSNR can be strongly influenced by large deviations in bright urban areas where radiance values are highest.

5.2 SSIM

The Structural Similarity Index (SSIM) evaluates similarity between images based on luminance, contrast, and structural patterns [20]. Unlike PSNR, SSIM attempts to approximate human visual perception by emphasizing structural relationships within the image.

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, \quad (4)$$

where μ_x and μ_y are the means of the two images, σ_x^2 and σ_y^2 are their variances, σ_{xy} is their covariance, and C_1, C_2 are small constants for numerical stability.

For nighttime light imagery, SSIM evaluates how well spatial lighting structures are preserved by the super-resolution reconstruction.

5.3 Radiance Error Metrics

Because NTL imagery represents physical measurements of radiance, direct error metrics are also considered. These metrics quantify how accurately reconstructed images preserve the magnitude of radiance values and are commonly used in remote sensing and geophysical data analysis.

Mean Absolute Error (MAE) MAE measures the average absolute difference between SR and HR radiance values.

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - x_i|. \quad (5)$$

MAE provides a straightforward measure of average radiance deviation and treats all errors linearly, making it relatively robust to outliers.

Root Mean Squared Error (RMSE) RMSE penalizes larger radiance errors more strongly than MAE.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - x_i)^2}. \quad (6)$$

Bias Bias measures the systematic difference between predicted and true radiance values, indicating whether the SR method tends to overestimate or underestimate brightness.

$$\text{Bias} = \frac{1}{N} \sum_{i=1}^N (y_i - x_i). \quad (7)$$

Bias is useful for identifying systematic radiometric shifts introduced by the super-resolution model, which may affect analyses relying on accurate radiance levels.

Pearson Correlation Pearson correlation measures the linear relationship between SR and HR radiance values across pixels.

$$r = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^N (y_i - \bar{y})^2}}. \quad (8)$$

A high Pearson correlation indicates that the spatial distribution of radiance values is preserved, even if absolute brightness levels differ.

Coefficient of Determination (R^2) The coefficient of determination represents the proportion of variance in HR radiance values that is explained by the SR reconstruction.

$$R^2 = 1 - \frac{\sum_{i=1}^N (x_i - y_i)^2}{\sum_{i=1}^N (x_i - \bar{x})^2}. \quad (9)$$

R^2 indicates how well the variability of radiance across the scene is captured by the super-resolved reconstruction.

By combining perceptual similarity metrics, radiometric error metrics, and correlation-based measures, the evaluation framework captures multiple dimensions of reconstruction quality: numerical accuracy, preservation of spatial lighting structures, and fidelity of radiometric relationships. Comparing the behavior of these metrics across different SR methods helps identify which measures provide the most informative evaluation of super-resolution performance for nighttime light satellite imagery.

6 Results

6.1 Metric Summary Statistics

Table 1: Summary statistics of reconstruction metrics for bicubic interpolation and the learned super-resolution model across the test dataset.

Metric	Method	Mean	Median	Std	Min	Max
Perceptual Similarity Metrics						
PSNR	Bicubic	31.30	30.80	5.96	19.77	80.00
PSNR	Learned SR	33.08	32.87	4.68	20.64	61.74
SSIM	Bicubic	0.821	0.857	0.109	0.457	0.999
SSIM	Learned SR	0.879	0.914	0.081	0.616	0.999
Radiometric Error Metrics						
RMSE	Bicubic	0.0309	0.0288	0.0140	0.0000	0.1027
RMSE	Learned SR	0.0251	0.0227	0.0130	-0.0008	0.0929
MAE	Bicubic	0.0197	0.0178	0.0107	0.0000	0.0798
MAE	Learned SR	0.0165	0.0142	0.0096	0.0002	0.0714
Bias	Bicubic	5.58×10^{-6}	1.07×10^{-6}	5.76×10^{-5}	-1.81×10^{-4}	4.26×10^{-4}
Bias	Learned SR	-5.58×10^{-4}	-4.04×10^{-4}	7.34×10^{-4}	-4.36×10^{-3}	6.96×10^{-4}
Correlation Metrics						
Pearson	Bicubic	0.914	0.958	0.121	0	0.992
Pearson	Learned SR	0.935	0.975	0.114	0	0.995
R^2	Bicubic	0.856	0.918	0.139	0.434	1
R^2	Learned SR	-1.08×10^8	0.949	1.08×10^9	-1.08×10^{10}	0.991

Higher values indicate better performance for PSNR, SSIM, Pearson and R^2 , while lower values indicate better performance for RMSE and MAE. For bias, values closer to zero indicate lower systematic radiometric error.

To provide an overview of reconstruction performance across the dataset, summary statistics were computed for each evaluation metric for both bicubic interpolation and the learned super-resolution model. For each metric, the mean, median, standard deviation, minimum, and maximum values were calculated across the 200 test images. These statistics provide a global view of reconstruction quality and allow comparison between the baseline interpolation method

and the learned model. Table 1 presents the summary statistics of all evaluation metrics across the test dataset for both bicubic interpolation and the learned super-resolution model.

For the perceptual reconstruction metrics, the learned model shows consistent improvements over bicubic interpolation. The mean PSNR increases from 31.30 dB for bicubic interpolation to 33.08 dB for the learned model, indicating improved signal fidelity. Similarly, the mean SSIM increases from 0.821 to 0.879 , suggesting that the learned model better preserves structural similarity with the reference high-resolution images. The standard deviation of these metrics is slightly lower for the learned model (PSNR std = 4.68 vs 5.96 , SSIM std = 0.081 vs 0.109), indicating somewhat more consistent performance across the dataset.

Radiometric error metrics show a similar trend. The mean RMSE decreases from 0.0309 for bicubic interpolation to 0.0251 for the learned model, while the mean MAE decreases from 0.0197 to 0.0165 . These reductions indicate that the learned model more accurately reconstructs the radiance values of the original high-resolution imagery. The bias for bicubic interpolation is close to zero (5.6×10^{-6}), whereas the learned model exhibits a small negative bias (-5.6×10^{-4}), suggesting a slight tendency to underestimate radiance values on average.

Correlation-based metrics also show moderate improvements for the learned model. The mean Pearson correlation increases from 0.914 for bicubic interpolation to 0.935 , indicating a stronger linear relationship between reconstructed and reference images. The mean R^2 for bicubic interpolation is 0.856 , suggesting that a large portion of the variance in the high-resolution data is explained by the reconstruction. However, the R^2 statistics for the learned model exhibit extreme variability, with large negative values in some cases, resulting in an unstable mean. This behavior likely arises in images with very low variance in radiance values, where the R^2 metric becomes numerically unstable.

Overall, the summary statistics indicate that the learned super-resolution model improves reconstruction quality across most evaluation metrics compared with bicubic interpolation. Improvements are observed in both perceptual similarity metrics (PSNR and SSIM) and radiometric error metrics (RMSE and MAE).

6.2 Correlation Between Evaluation Metrics

Correlation matrices were computed for the learned SR method using both the Pearson Correlation Coefficient and the Spearman Rank Correlation Coefficient. These matrices quantify the strength of relationships between metrics across all evaluated test images. The results are visualized in the correlation heatmaps shown in Figure 1 and Figure 2, where stronger correlations are indicated by values closer to ± 1 . The objective is to determine whether certain metrics capture similar aspects of reconstruction quality or whether they describe different characteristics of the super-resolved nighttime lights imagery.

6.2.1 Overall Correlation Structure

The correlation matrices reveal strong relationships among several commonly used image quality metrics. In particular, Peak Signal-to-Noise Ratio, Structural Similarity Index Measure, Root Mean Square Error, and Mean Absolute Error show high correlations across the dataset (Figure 1 and Figure 2).

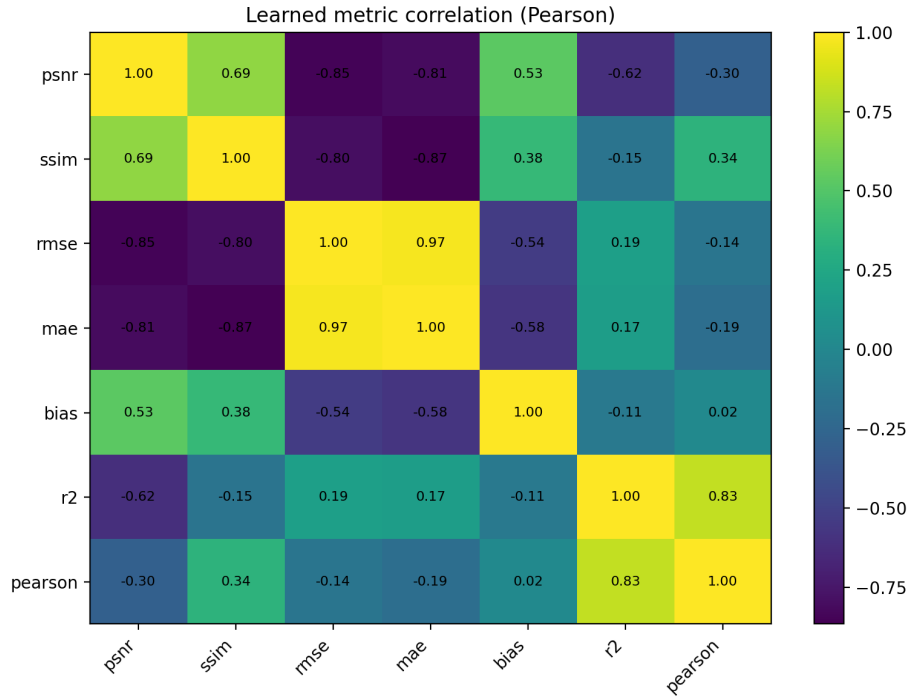


Figure 1: Pearson correlation matrix between evaluation metrics for the learned super-resolution model across all test images. Values close to 1 or -1 indicate strong positive or negative relationships between metrics, respectively.

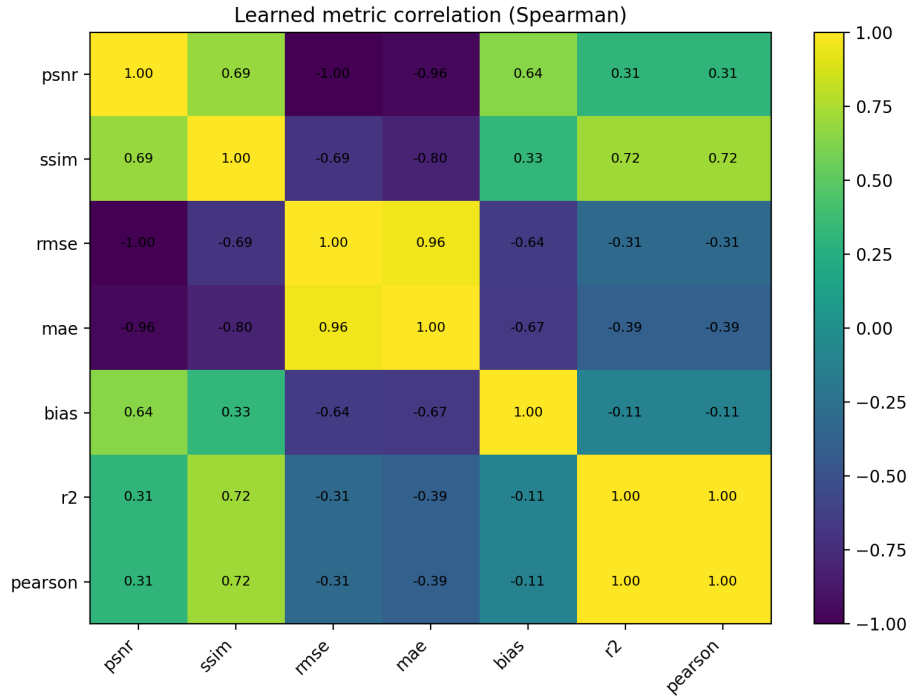


Figure 2: Spearman correlation matrix between evaluation metrics for the learned super-resolution model across all test images. Values close to 1 or -1 indicate strong positive or negative relationships between metrics, respectively.

As expected, metrics where higher values indicate better performance (PSNR and SSIM) show strong negative correlations with metrics where lower values indicate better performance (RMSE and MAE). This relationship arises because all four metrics are closely linked to the magnitude of pixel-wise differences between the reconstructed and reference images. For example, PSNR is mathematically derived from the mean squared error and therefore shows particularly strong correlation with RMSE. Similarly, SSIM tends to increase as pixel-level errors decrease, resulting in strong associations with both PSNR and the error-based metrics.

6.2.2 Radiometric Relationship Metrics

In addition to the image quality metrics above, several radiometric metrics were computed, including bias, the Coefficient of Determination, and the Pearson Correlation Coefficient between predicted and reference radiance values.

Bias measures the systematic over- or underestimation of radiance values and therefore does not necessarily follow the same patterns as the error-based metrics. As shown in the correlation matrices (Figure 1 and Figure 2), bias generally exhibits weaker correlations with PSNR, SSIM, RMSE, and MAE. This indicates that images with low pixel-wise reconstruction error can still exhibit small systematic shifts in brightness.

By contrast, the radiance-based correlation metrics (R^2 and Pearson radiance correlation) show strong positive correlation with each other, since both measure how well the spatial variability of radiance values is preserved between the reconstructed and reference images.

6.2.3 Pearson and Spearman Comparison

The Pearson and Spearman correlation matrices show similar overall patterns (Figure 1 and Figure 2). Because Pearson correlation measures linear relationships, while Spearman correlation evaluates rank-based monotonic relationships, the similarity between the two matrices suggests that the relationships between most metrics are largely monotonic and not strongly influenced by outliers within the dataset.

6.2.4 Comparison of Metric Correlations Across Reconstruction Methods

The Pearson correlation matrices for bicubic interpolation and the learned super-resolution model show broadly similar relationships between the evaluation metrics. In both cases, Root Mean Square Error and Mean Absolute Error are extremely strongly correlated, while Peak Signal-to-Noise Ratio and Structural Similarity Index Measure show strong negative correlations with the error metrics. A key difference appears in the radiometric relationship metrics: the correlation between Coefficient of Determination and the radiance Pearson Correlation Coefficient is stronger for the learned model than for bicubic interpolation. Bias remains weakly correlated with the other metrics in both cases.

6.3 Comparison of Super-Resolution and Bicubic Interpolation

To assess both the effectiveness of the learned super-resolution (SR) model and the behavior of the evaluation metrics, the learned SR reconstruction was compared with bicubic interpolation

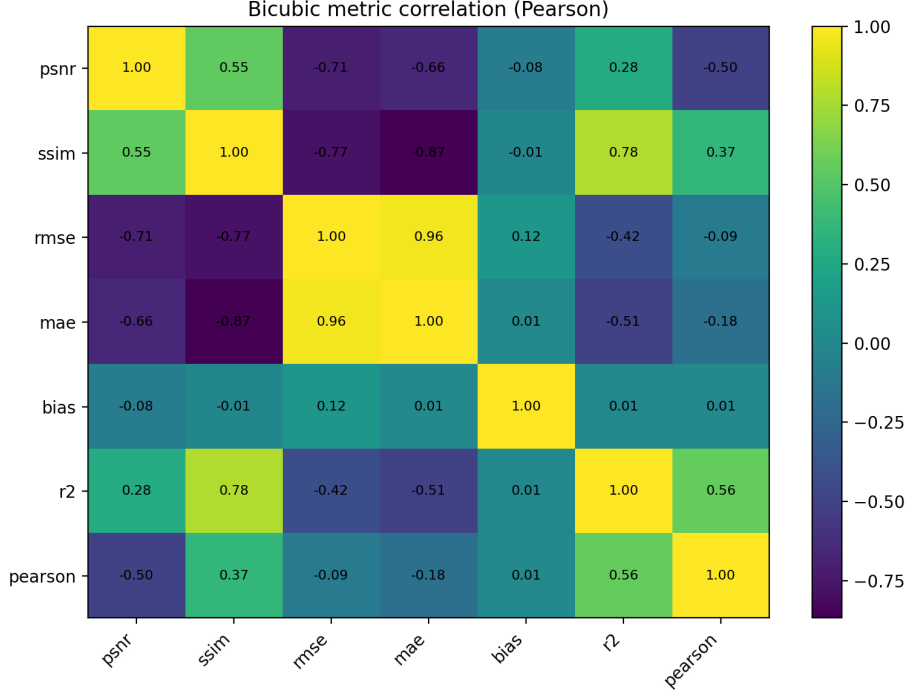


Figure 3: Pearson correlation matrix between evaluation metrics for the bicubic super-resolution model across all test images. Values close to 1 or -1 indicate strong positive or negative relationships between metrics, respectively.

across all test images. Bicubic interpolation serves as a standard baseline in super-resolution tasks because it performs deterministic spatial interpolation without learning from the data. This comparison therefore provides a useful reference for interpreting how different metrics respond to improvements in reconstruction quality. Figure 4 shows the normalized mean values of all evaluation metrics for both methods. Across most metrics, the learned SR model performs better than bicubic interpolation. In particular, PSNR increases while RMSE and MAE decrease, indicating reduced pixel-level reconstruction error. At the same time, higher SSIM values suggest that the learned model better preserves structural patterns present in the nighttime light imagery.

In contrast, bias values for the bicubic method are better than those of the learner SR method. This indicates that the SR method introduces a radiometric shift.

Visual comparisons between bicubic and learned SR outputs (Figure 5 and Figure 6) support these quantitative findings. The learned SR reconstructions exhibit sharper spatial structures and more coherent light patterns, particularly in urban areas where fine-scale illumination gradients are present, while bicubic interpolation produces smoother and slightly blurred representations.

Overall, the comparison demonstrates that the learned SR model improves reconstruction quality relative to bicubic interpolation. At the same time, the results provide insight into the relative usefulness of different evaluation metrics, showing that pixel-based error metrics (PSNR, RMSE, MAE) and structural metrics (SSIM) are effective at capturing these improvements, while bias provides limited discriminatory power in this context.

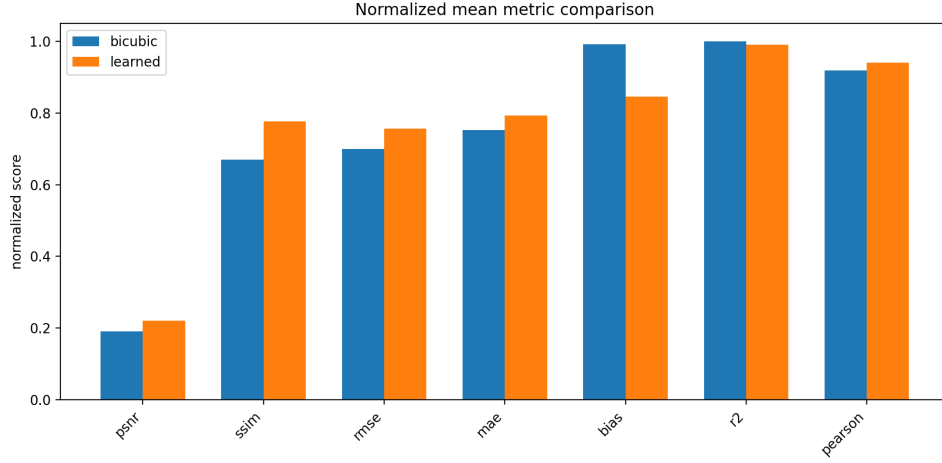


Figure 4: Normalized mean values of evaluation metrics for bicubic interpolation and the learned super-resolution model across the test dataset. Metrics are normalized to allow comparison across different scales and directions, highlighting the relative performance of both methods.

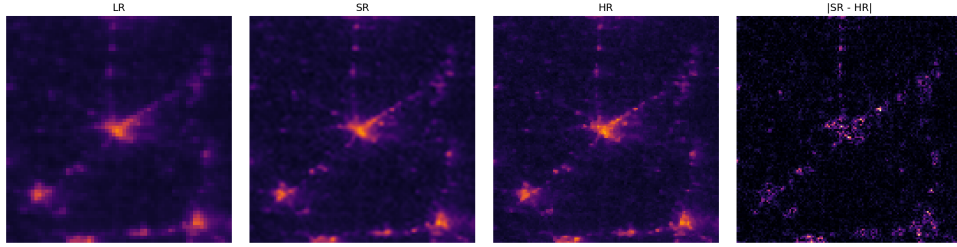


Figure 5: Example NTL chip of Paris showing the low-resolution input (LR), learned super-resolution output (SR), ground truth high-resolution image (HR), and the absolute error map $|SR - HR|$.

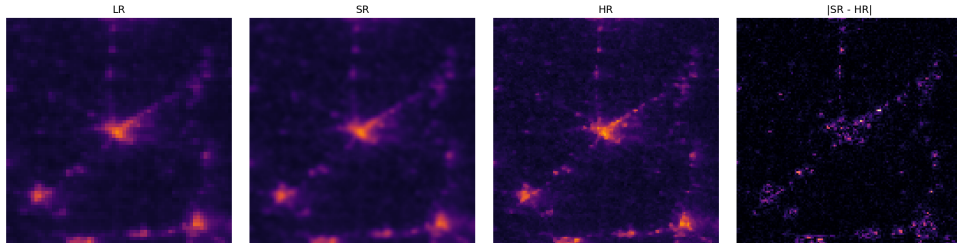


Figure 6: Example NTL chip of Paris showing the low-resolution input (LR), bicubic super-resolution output (SR), ground truth high-resolution image (HR), and the absolute error map $|SR - HR|$.

6.4 Image-Level Analysis of Metric Behavior

To better understand how evaluation metrics behave across different scenes for the learned SR, the top- and worst-ranked images for each metric were analyzed. For each metric, the ten best and ten worst samples were identified based on the per-image results. Examining these cases allows us to identify systematic conditions under which certain metrics succeed or fail, and therefore assess their suitability for evaluating night-time lights (NTL) super-resolution reconstruction.

Because NTL imagery exhibits strong spatial and radiometric variability, ranging from dense urban light clusters to sparsely illuminated rural regions, the behaviour of reconstruction metrics can differ substantially across scenes. The analysis below highlights typical patterns observed in the ranking results.

6.4.1 Structural Metrics: SSIM

The images with the highest SSIM scores generally correspond to scenes where the spatial organization of night-time lights forms coherent linear or clustered structures, such as road networks, elongated settlements, or connected urban corridors. In these cases, the learned super-resolution model successfully reconstructs the relative spatial arrangement of light sources, even when individual light intensities differ slightly from the ground truth. Several of the top-ranked samples exhibit elongated lighting patterns following transportation infrastructure or settlement corridors. For example, the reconstruction shown in NTL Chip Astrakhan 7 preserves the linear structure of the illuminated corridor and the relative spacing between light clusters. While the low-resolution input shows blurred light distributions, the super-resolution output restores sharper spatial separation between individual light sources and closely matches the structure visible in the high-resolution reference.

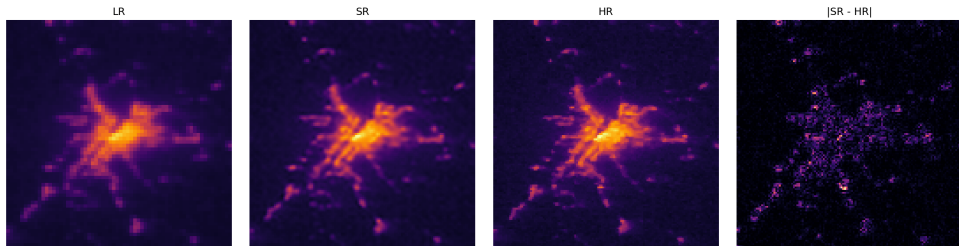


Figure 7: NTL Chip Astrakhan, learned SR output

A similar behaviour can be observed in NTL Chip Cádiz 8 and NTL Chip Hyères 9, where the reconstructed image maintains the geometry of extended lighting structures across the scene. In these cases, the model successfully enhances spatial detail while preserving the overall morphology of the illuminated regions. The corresponding difference maps show that most reconstruction errors are distributed across low-intensity areas rather than concentrated within the main structural features.

Interestingly, several high-SSIM examples correspond to moderate-intensity lighting networks rather than extremely bright urban cores. This can be observed in NTL Chip Ziyodin Shaharchasi 10 and NTL Chip Krasnoarmeysk 11, where the scenes contain multiple small light clusters

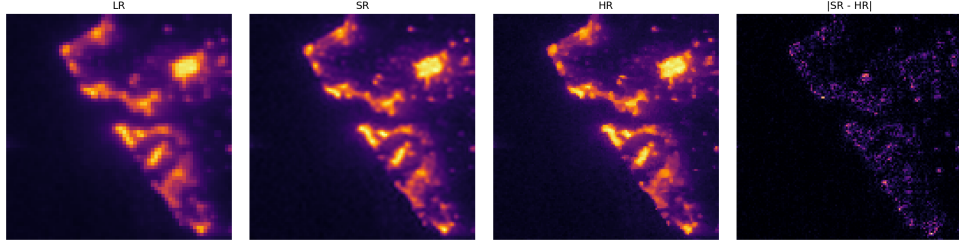


Figure 8: NTL Chip Cádiz, learned SR output

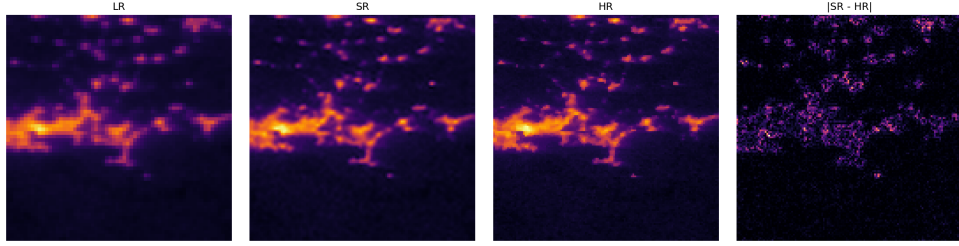


Figure 9: NTL Chip Hyères, learned SR output

connected by faint linear structures. The super-resolution model reconstructs these patterns well, leading to strong structural similarity despite minor radiometric differences. Across the dataset, many scenes with SSIM values above approximately 0.95 share this characteristic, containing clearly defined lighting networks with moderate radiance levels that allow the model to recover spatial structure without strong saturation effects.

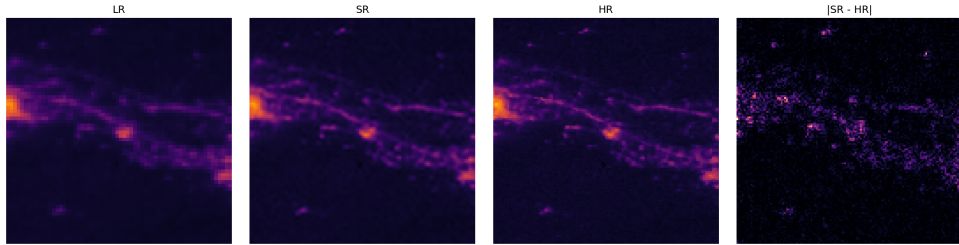


Figure 10: NTL Chip Ziyodin Shaharchasi, learned SR output

At the same time, the analysis also reveals that very high SSIM values can occur in scenes with extremely low overall radiance. For example, NTL Chip Savissivik [12](#) achieves an SSIM value close to 1 despite containing almost no visible lighting structures. In such cases, both the super-resolution output and the ground truth consist largely of dark pixels with minimal spatial variation. Because SSIM compares local image statistics such as mean intensity and variance, scenes dominated by near-zero radiance can produce artificially high similarity scores even when the reconstruction contains little meaningful structural information.

In contrast, scenes dominated by sparse illumination combined with diffuse background radiance or sensor noise tend to produce lower SSIM scores. The lowest-ranked samples, such as NTL Chips Akpassi [13](#), and Segala Mba [14](#), contain extremely weak and spatially diffuse radiance signals with few clearly defined light sources. In these cases, the super-resolution output introduces additional fine-scale texture across the image, leading to differences in local statistics

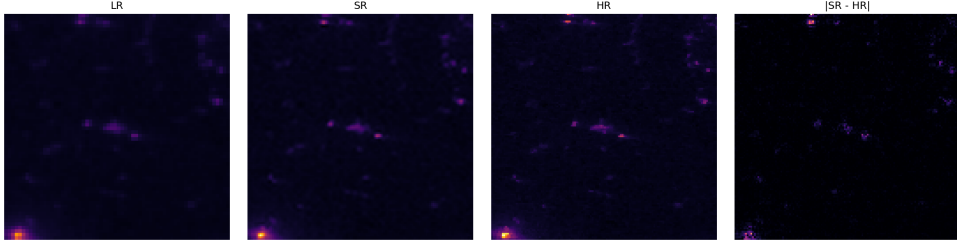


Figure 11: NTL Chip Krasnoarmeysk, learned SR output

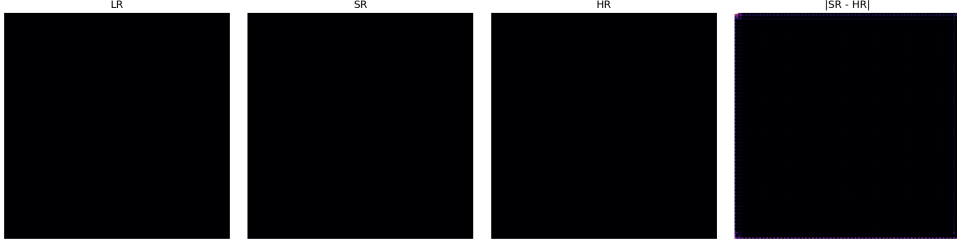


Figure 12: NTL Chip Savissivik, learned SR output

relative to the ground truth. The corresponding difference maps show errors distributed broadly across the scene rather than concentrated around specific lights, which reduces the structural similarity measured by SSIM.

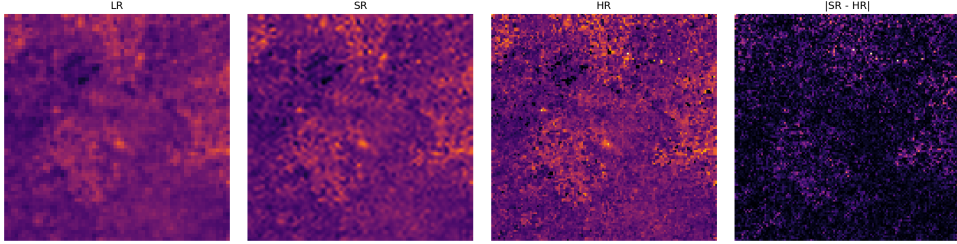


Figure 13: NTL Chip Akpassi, learned SR output

Overall, the image-level analysis suggests that SSIM performs particularly well for NTL scenes containing coherent spatial lighting structures with moderate radiance levels, such as road networks or distributed settlements. However, the metric becomes less reliable in scenes dominated by very weak signals or diffuse background noise, and it may also produce artificially high scores in near-dark scenes with minimal spatial variation. These observations highlight that while SSIM captures spatial structure effectively, it does not always reflect the radiometric fidelity of reconstructed night-time light intensities.

6.4.2 Pixel-Error Metrics: RMSE and MAE

The behaviour of RMSE and MAE is strongly influenced by the absolute radiance values present in the scene. Images with the lowest (best) RMSE and MAE scores generally correspond to scenes with very weak or moderately distributed illumination, where most pixels have near-zero radiance. In these cases, the super-resolution model produces reconstructions that closely match the overall intensity distribution of the ground truth. For example, NTL Chip Savissivik [12](#)

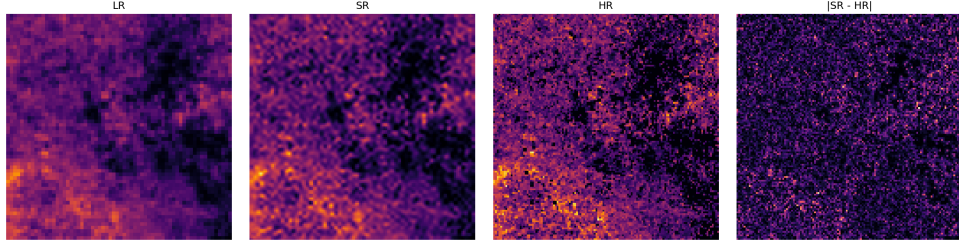


Figure 14: NTL Chip Segala Mba, learned SR output

exhibits extremely low RMSE and MAE values because both the prediction and reference image consist almost entirely of dark pixels with only minimal signal.

Several other low-error samples correspond to small settlements embedded in largely dark rural regions. Examples include NTL Chip Douar Ras el Ksar [15](#), and NTL Chip São Miguel do Araguaia [16](#), where only a few localized light sources are present. In these cases, most pixels contribute little to the error because their radiance is near zero, allowing the model to achieve favorable RMSE and MAE scores even when small deviations occur around the illuminated pixels.

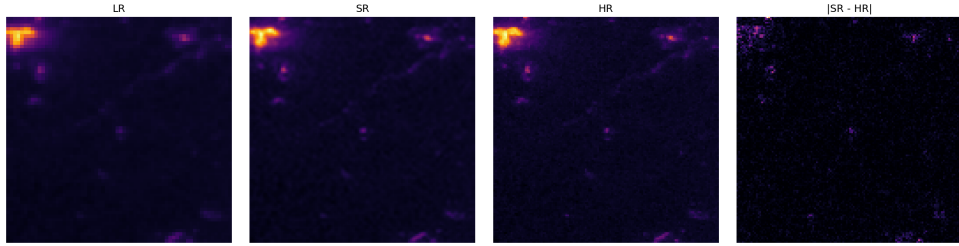


Figure 15: NTL Chip Douar Ras el Ksar, learned SR output

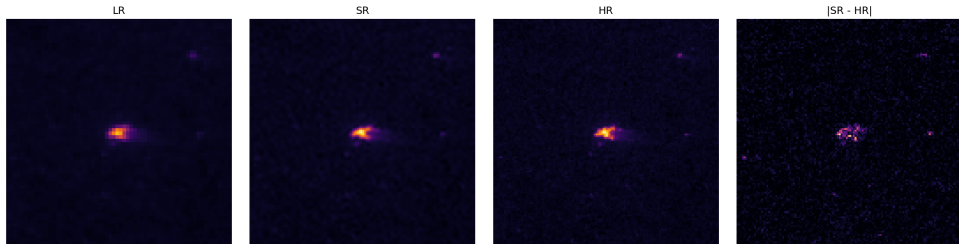


Figure 16: NTL Chip São Miguel do Araguaia, learned SR output

However, this behaviour highlights an important limitation of pixel-error metrics for night-time light imagery. Because RMSE and MAE average errors across all pixels, scenes dominated by dark background areas can receive very strong scores even when the reconstruction does not perfectly capture small isolated light sources. For example, in NTL Chip Tinogasta [17](#), the reconstruction slightly smooths the spatial extent of the main light source, yet the overall error remains low due to the dominance of low-intensity pixels across the scene.

The highest RMSE and MAE values, in contrast, typically occur in scenes containing noisy low-intensity signals or spatially diffuse radiance patterns rather than bright urban centers. Examples include NTL Chips Segala Mba [14](#), and Akpassi [13](#), where the ground truth contains

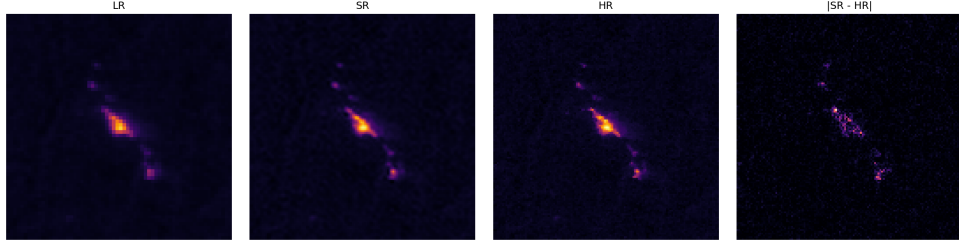


Figure 17: NTL Chip Tinogasta, learned SR output

scattered faint lights combined with background radiometric variability. In these cases, the super-resolution model introduces additional fine-scale texture across the image, producing widespread pixel-level differences relative to the reference image. The corresponding difference maps show errors distributed across large portions of the scene rather than concentrated around specific light sources.

Another pattern observed among high-error samples is the presence of very small, isolated light sources surrounded by weak radiance gradients, as seen in NTL Chip Mao [18](#) and NTL Chip Takieta [19](#). Because RMSE penalizes intensity differences across all pixels, even subtle variations in the surrounding radiance field can accumulate into relatively large aggregate errors.

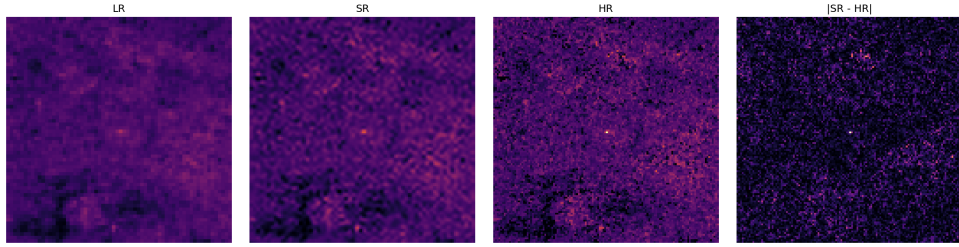


Figure 18: NTL Chip Mao, learned SR output

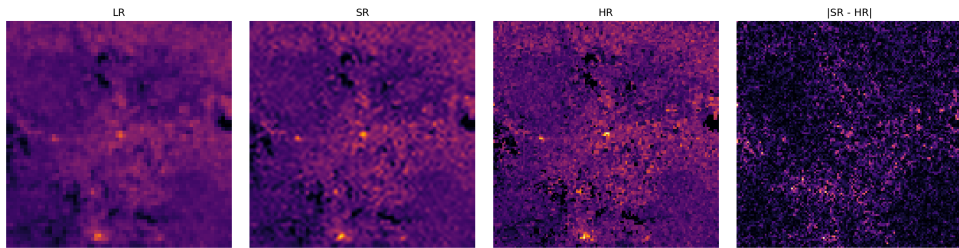


Figure 19: NTL Chip Takieta, learned SR output

Overall, the image-level analysis indicates that RMSE and MAE are strongly dependent on the global radiometric distribution of the scene. The metrics tend to produce very favorable scores for images dominated by dark background pixels, while scenes containing diffuse low-intensity signals or noisy radiance patterns can receive substantially higher error values. These observations suggest that pixel-error metrics primarily reflect radiometric differences in overall intensity, and may not fully capture whether the spatial structure of night-time lighting has been accurately reconstructed.

6.4.3 Signal Fidelity Metric: PSNR

The ranking of images by PSNR exhibits patterns similar to those observed for RMSE, since PSNR is directly derived from the mean squared reconstruction error. High PSNR scores typically occur in scenes where the predicted image is numerically very close to the ground truth across most pixels, particularly in areas dominated by low radiance values.

Across the dataset, many images with PSNR values above approximately 35 dB share a similar structure: they contain only a few isolated light sources embedded in large dark background regions. For example, in NTL Chips Canarana I [20](#) and Tinogasta [17](#), where a single dominant light cluster occupies only a small fraction of the scene. In these cases the super-resolution output remains numerically close to the reference image across most pixels, even though the reconstructed light cluster is still somewhat smooth and lacks the sharper spatial detail expected from super-resolution.

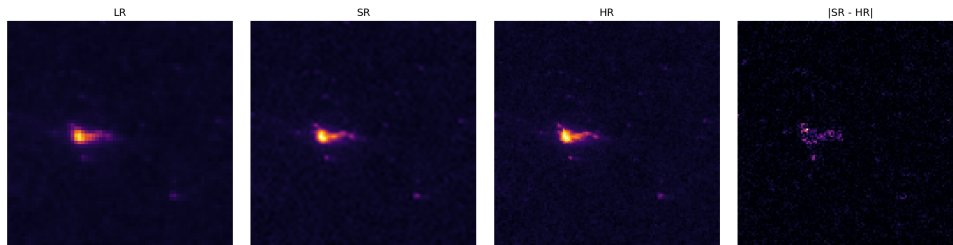


Figure 20: NTL Chip Canarana I, learned SR output

In scenes containing more complex lighting networks, PSNR values are often slightly lower despite visually meaningful improvements. For example, NTL Chip Krasnoarmeysk [11](#) contains multiple neighbouring light sources and more heterogeneous radiance patterns. The super-resolution reconstruction improves the separation between these light clusters relative to the low-resolution input, but small radiometric deviations within the brighter regions increase the squared error and therefore reduce the PSNR value.

The lowest PSNR scores are typically associated with scenes dominated by diffuse radiance patterns and background noise, where the reconstruction struggles to reproduce the fine-scale radiometric variability of the ground truth. Examples include NTL Chips Akpassi [13](#), and Segala Mba [14](#), where the ground truth images contain spatially heterogeneous radiance fields with many weak scattered signals. In these cases the super-resolution output introduces additional texture across the image, producing widespread pixel-level differences relative to the reference image, as visible in the difference maps.

A similar effect can be observed in NTL Chips Mao [18](#), and Nang Rong [21](#), where large portions of the image contain noisy low-intensity radiance rather than clearly defined light clusters. Because PSNR penalizes squared intensity differences across all pixels, these distributed radiometric deviations lead to substantially lower scores compared to scenes with simpler lighting structures.

Overall, the image-level analysis suggests that PSNR primarily reflects global radiometric similarity rather than structural reconstruction quality. Scenes dominated by dark backgrounds or simple lighting patterns tend to achieve the highest PSNR values, while scenes with complex radiance distributions or diffuse low-intensity signals receive lower scores even when the recon-

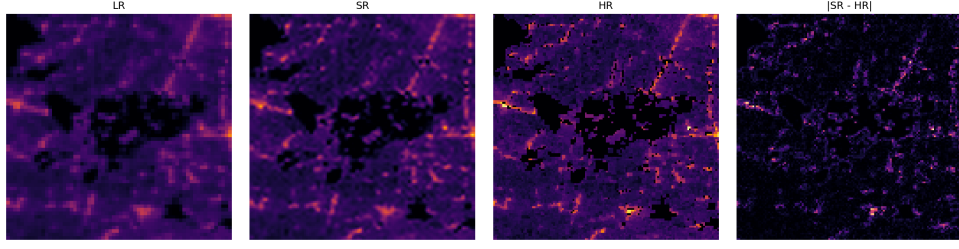


Figure 21: NTL Chip Nang Rong, learned SR output

structed spatial structure appears visually reasonable. This indicates that PSNR alone may not reliably capture improvements in the spatial organization of night-time light patterns.

6.4.4 Radiometric Consistency: Bias

The bias metric measures the mean signed difference between predicted and ground-truth radiance and therefore provides insight into systematic over- or under-estimation of radiance values by the model. In contrast to pixel-error metrics, bias reflects whether reconstructed radiance values tend to be consistently shifted relative to the reference image.

Images with bias values close to zero generally correspond to scenes where positive and negative reconstruction errors are spatially balanced across the image. This behaviour can be observed in NTL Chips Visby [22](#) and Simav [23](#), where the reconstructed settlement patterns closely follow the spatial distribution of the ground truth. Although small deviations occur around individual light clusters, the difference maps show that over- and under-estimation occur in different parts of the scene, resulting in minimal net bias.

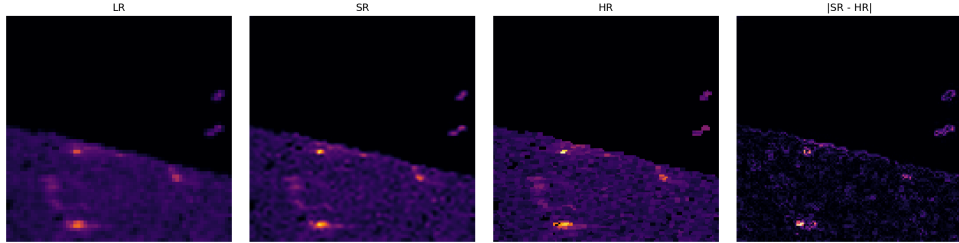


Figure 22: NTL Chip Visby, learned SR output

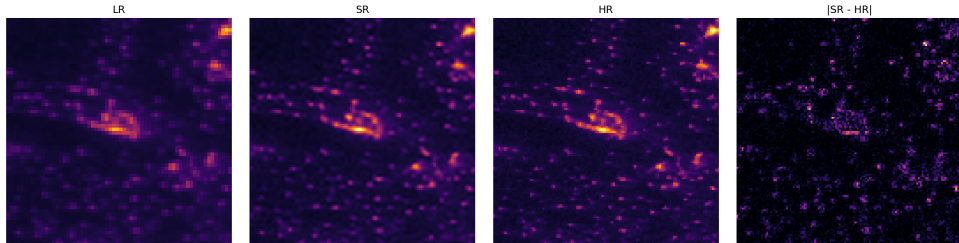


Figure 23: NTL Chip Simav, learned SR output

A similar pattern can be observed in NTL Chips Plouzané [24](#) and Tianmingsi [25](#), which contain elongated lighting structures and moderately bright settlements. In these cases, the super-resolution reconstruction slightly sharpens the spatial organization of the lights while

maintaining a radiance distribution that remains very similar to the reference image. Because small radiometric deviations occur in both directions around the illuminated areas, the resulting bias remains close to zero.

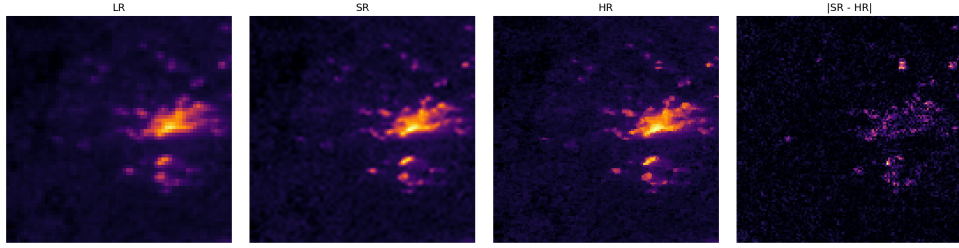


Figure 24: NTL Chip Plouzané, learned SR output

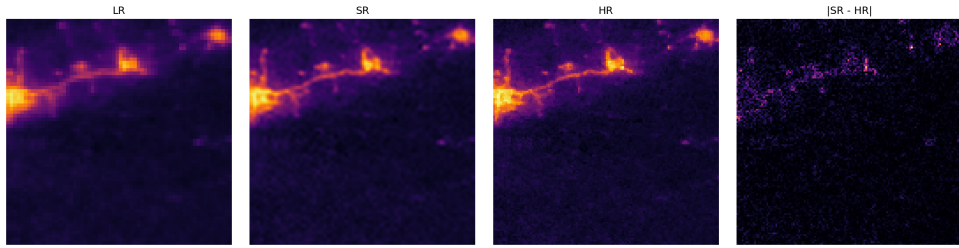


Figure 25: NTL Chip Tianmingsi, learned SR output

In contrast, larger bias values tend to occur in scenes containing very bright light clusters or highly heterogeneous radiance patterns. For example, in NTL Chip Wilrijk [26](#) and NTL Chip Higashi-Matsuyama [27](#), the brightest urban cores appear somewhat attenuated in the reconstructed output relative to the ground truth. The corresponding difference maps show concentrated deviations around these high-intensity regions, indicating that the model slightly underestimates the peak radiance of the brightest lights.

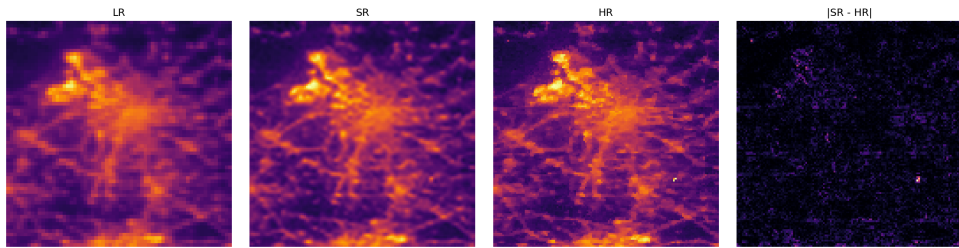


Figure 26: NTL Chip Wilrijk, learned SR output

Positive bias values can also occur in scenes dominated by diffuse background radiance and weak, spatially distributed signals. In such cases, the super-resolution reconstruction tends to produce slightly higher low-intensity values across the image compared to the reference image. This behaviour can be observed in NTL Chips Akpassi [13](#) and Bedourie [28](#), where the reconstructed output exhibits a smoother but slightly brighter background radiance field than the ground truth. While the differences are subtle at the pixel level, they appear across large portions of the image, as visible in the difference maps. The accumulation of these small

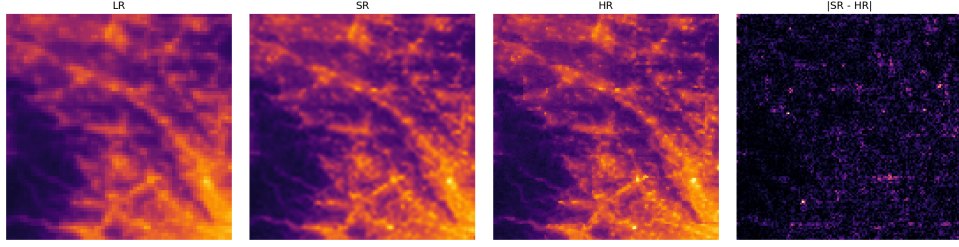


Figure 27: NTL Chip Higashi-Matsuyama, learned SR output

positive deviations across many pixels leads to a measurable positive shift in the average radiance, resulting in a positive bias value for these scenes.

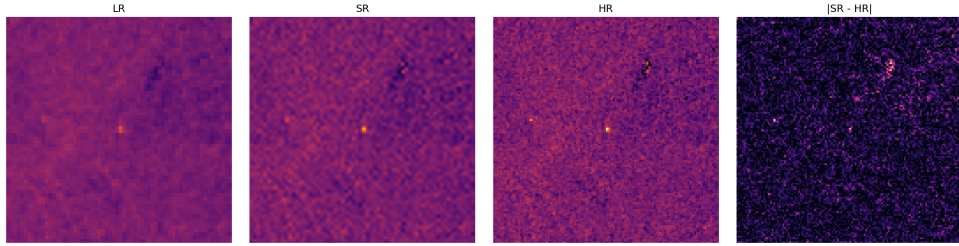


Figure 28: NTL Chip Bedourie, learned SR output

Overall, the image-level analysis suggests that bias primarily reflects systematic radiometric shifts within specific intensity regimes. Scenes containing bright urban cores tend to produce negative bias due to slight attenuation of peak intensities, whereas scenes dominated by weak signals may exhibit positive bias when faint background illumination is introduced. As a result, bias is most informative when interpreted together with spatial and pixel-error metrics that reveal how these radiometric deviations are distributed across the scene.

6.4.5 Linear Radiometric Agreement: R^2 and Pearson Correlation:

The Pearson correlation coefficient and the coefficient of determination (R^2) measure the degree of linear agreement between predicted and ground-truth radiance values across all pixels within an image. Unlike pixel-error metrics such as RMSE or MAE, which quantify absolute radiometric differences, these metrics evaluate whether the relative ordering of pixel intensities is preserved by the reconstruction.

Images with the highest Pearson and R^2 values generally correspond to scenes where the overall spatial distribution of radiance intensities is well preserved. This behaviour can be observed in NTL Chips Cádiz 8, and Pataskala 29, where the reconstructed images closely reproduce complex urban lighting networks visible in the ground truth. In these scenes, bright urban cores, surrounding road networks, and gradually decreasing radiance toward darker areas maintain consistent intensity relationships between the reconstructed and reference images.

A similar pattern can be observed in NTL Chips Granadero Baigorria 30 and Cantonment 31, which contain branching lighting corridors extending from a central urban core. The super-resolution reconstruction preserves the main lighting structures and relative brightness gradients across the scene, resulting in strong linear agreement between predicted and ground-truth radiance

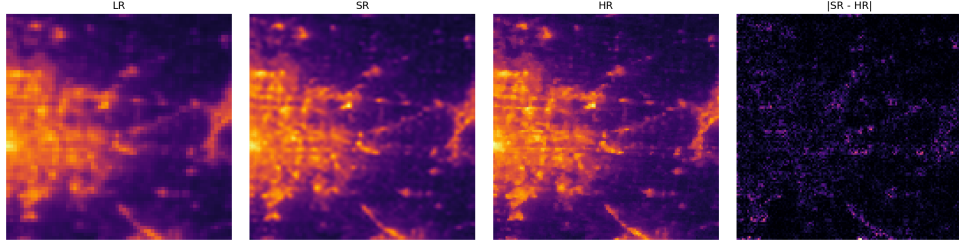


Figure 29: NTL Chip Pataskala, learned SR output

values. High correlation values are also observed in scenes containing multiple distributed light clusters, such as NTL Chip Johnston [32](#), where the model maintains the relative intensity differences between settlements and surrounding dark regions.

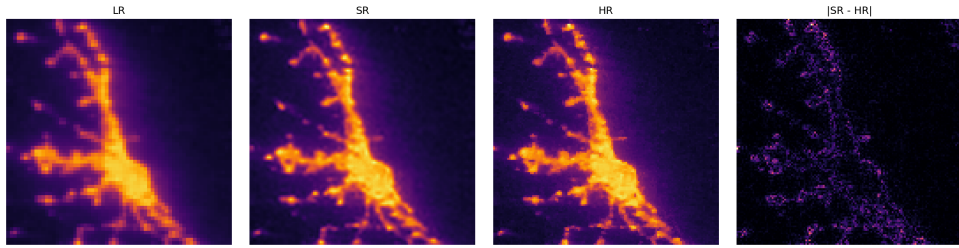


Figure 30: NTL Chip Granadero Baigorria, learned SR output

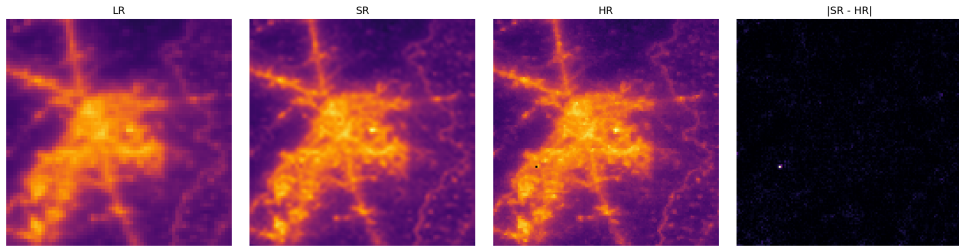


Figure 31: NTL Chip Cantonment, learned SR output

In contrast, the lowest Pearson values typically occur in scenes dominated by diffuse background radiance or noisy low-intensity signals, such as in NTL Chip Akpassi [13](#). In such cases, small variations in the reconstructed background radiance alter the relative ordering of pixel intensities across the image, reducing the linear relationship between predicted and reference values. Because Pearson correlation depends on consistent radiance relationships across pixels, these distributed fluctuations can substantially reduce the measured correlation even when the reconstruction appears visually similar.

Overall, the analysis suggests that Pearson correlation and R^2 primarily capture whether the global radiance structure of a scene is preserved, rather than the magnitude of local reconstruction errors. Scenes with well-defined urban lighting networks or stable brightness gradients tend to achieve the highest scores, while scenes dominated by weak or noisy radiance signals produce lower correlation values.

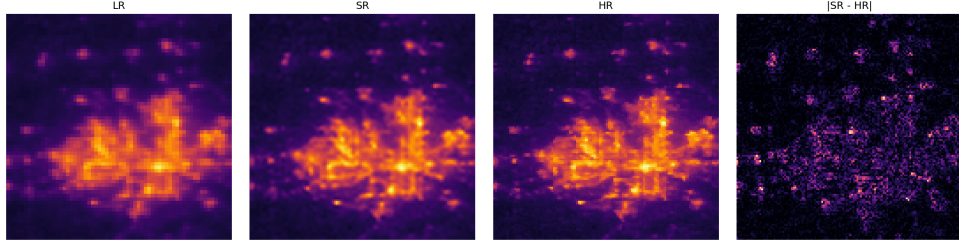


Figure 32: NTL Chip Johnston, learned SR output

6.5 Implications for Metric Selection

The results show that metric selection has a substantial influence on how reconstruction quality is interpreted for nighttime lights (NTL) super-resolution. Because NTL imagery combines large dark background areas, sparse weak signals, complex urban light networks, and occasionally saturated bright cores, different metrics respond to different properties of the reconstruction. A reliable evaluation therefore requires a small set of complementary metrics rather than a single summary score.

First, the strong correlations between PSNR, RMSE, and MAE indicate that these metrics describe closely related aspects of performance, namely pixel-wise radiometric error. This is also consistent with the image-level analysis, where all three metrics tend to favour scenes dominated by dark background pixels and relatively simple radiance distributions. In such cases, a reconstruction can achieve very good error scores even if small light sources remain slightly blurred or spatial detail is not fully restored. For this reason, using all three metrics together provides only limited additional information. In practice, RMSE or MAE is sufficient as a direct error measure, while PSNR may be retained as a standard image-reconstruction metric for easier comparison with related super-resolution literature.

Second, SSIM provides important complementary information because it is more sensitive to the spatial organisation of illuminated structures. The image-level results show that SSIM is particularly informative for scenes containing coherent lighting patterns such as settlement corridors, road networks, or distributed urban clusters, where structural preservation is a central objective of NTL reconstruction. At the same time, SSIM must be interpreted with care in near-dark scenes, where very high scores can arise simply because both prediction and reference contain little spatial variation. SSIM is therefore valuable, but it should not be used in isolation.

Third, Pearson correlation contributes a different perspective by measuring whether the relative ordering of radiance values is preserved across the image. This is especially relevant for NTL data, where maintaining the broader radiometric structure of a scene can matter as much as minimizing absolute pixel error. The results suggest that Pearson correlation is most informative in scenes with clear brightness gradients or structured urban radiance fields, and it complements RMSE/MAE by capturing agreement in radiance relationships rather than only absolute deviations.

By contrast, R^2 appears less suitable as a primary evaluation metric in this setting. Although it is conceptually related to Pearson correlation and can be informative in stable scenes, the summary statistics show that it becomes numerically unstable in low-variance images, producing extreme negative values and an unreliable mean. Since low-variance and near-dark scenes are

common in NTL imagery, this instability limits the usefulness of R^2 for aggregate comparison across the dataset.

Bias plays a distinct role among the evaluated metrics. The correlation analysis shows that bias is only weakly related to the other metrics and therefore captures a different aspect of reconstruction quality. While it provides limited discriminatory power when comparing bicubic interpolation and the learned SR model, it directly reflects systematic over- or underestimation of radiance values. This property is particularly relevant for remote sensing applications of nighttime lights imagery, where absolute radiance values are often used as indicators of human activity, infrastructure usage, or socio-economic conditions. Systematic radiometric shifts introduced by a reconstruction model could therefore lead to misleading interpretations when analyzing temporal changes in lighting patterns. For this reason, bias remains an important metric for evaluating the radiometric consistency of reconstructed NTL data, even if it is less informative for distinguishing overall reconstruction quality.

Taken together, the results suggest that evaluating NTL super-resolution requires a combination of complementary metrics capturing different properties of the reconstruction. A practical evaluation framework should therefore include one metric for absolute radiometric error (RMSE or MAE), one metric for spatial structure (SSIM), and one metric for preservation of radiance relationships (Pearson correlation). In addition, bias should be reported to assess systematic radiometric shifts that may affect downstream remote sensing analyses relying on accurate radiance values. PSNR may be included for comparability with conventional image super-resolution studies, while the use of R^2 should be treated with caution due to its numerical instability in low-variance NTL scenes. Together, this set of metrics provides a balanced evaluation of radiometric accuracy, spatial reconstruction quality, and preservation of meaningful radiance patterns.

7 Limitations and Future Work

Several limitations of the present study should be considered when interpreting the results and identifying directions for future research.

First, the evaluation was conducted on a dataset of approximately 2000 night-time light image chips of cities. Although the dataset contains a range of scenes, its geographic coverage remains limited and is biased toward regions with illumination. Urban areas typically contain more spatial structure and higher radiance values, which can favor metrics such as SSIM and PSNR. Consequently, the observed metric behavior may not fully represent performance in sparsely lit rural regions or in areas with extremely low radiance levels. Expanding the dataset to include a larger and more geographically diverse set of samples would improve the robustness of the conclusions.

Second, the analysis relies on a single learned super-resolution model and a specific down-sampling procedure used to generate the low-resolution inputs. Different super-resolution architectures can produce distinct reconstruction characteristics, such as sharper edges, smoother textures, or hallucinated details. As a result, the behavior and ranking of evaluation metrics may depend partly on the specific model and degradation process considered. Future work could evaluate metric performance across multiple super-resolution models and degradation settings, or

investigate whether certain metrics are better suited to specific classes of reconstruction methods.

Another limitation concerns the evaluation of radiometric fidelity. While error-based metrics such as RMSE, MAE, and bias are included, the analysis focuses primarily on statistical metric behavior rather than on the implications for downstream remote sensing applications. In night-time light data, accurate radiance estimation is essential for applications such as monitoring economic activity, disaster impacts, or infrastructure development. Future work could therefore assess how reconstruction errors propagate into application-oriented analyses and whether additional calibration steps are required to preserve radiometric consistency.

Several directions for future work follow from these limitations. A natural extension would be to evaluate metric behavior across multiple super-resolution models, including architectures with different reconstruction properties. Such analysis could also explore whether evaluation metrics should be tailored to the characteristics of the specific reconstruction model being used. In addition, future research could focus on radiometric calibration and application-oriented evaluation, examining how super-resolution outputs influence downstream tasks such as change detection or disaster assessment.

Further work could also investigate the temporal behavior of super-resolved night-time light imagery. Since many applications rely on time-series observations, it is important to assess whether super-resolution methods preserve consistent radiance patterns across time. Finally, the development of evaluation metrics specifically designed for night-time light imagery represents an important research direction. Metrics that explicitly account for sparse illumination patterns, high dynamic range, and radiometric accuracy may provide more reliable evaluation than traditional image quality metrics originally developed for natural images.

8 Conclusion

This project investigated the evaluation of super-resolution methods for nighttime light (NTL) satellite imagery, with a particular focus on the suitability of different quantitative metrics. Using VIIRS-based NTL image chips, a learned super-resolution model was compared with bicubic interpolation, and reconstruction quality was analyzed through perceptual, error-based, and radiometric metrics.

The results show that the learned model improves reconstruction quality over bicubic interpolation across most metrics, particularly PSNR, SSIM, RMSE, MAE, and Pearson correlation. At the same time, the analysis demonstrates that these metrics do not provide equivalent information. PSNR, RMSE, and MAE largely reflect pixel-wise radiometric error, SSIM is more sensitive to the preservation of spatial structure, and Pearson correlation captures how well radiance relationships are maintained across the scene. Bias, while less effective for distinguishing overall reconstruction quality, remains important for identifying systematic radiometric over- or underestimation, which is especially relevant in remote sensing applications where absolute radiance values carry physical meaning.

Overall, the findings suggest that evaluating NTL super-resolution requires a combination of complementary metrics rather than reliance on a single score. A practical evaluation framework should therefore include an absolute error metric, a structural similarity metric, and a radiometric relationship metric, while also reporting bias to assess radiometric consistency. More broadly,

the study highlights that metric choice has a direct impact on how reconstruction quality is interpreted for NTL imagery, and that evaluation strategies for this domain should account for both visual structure and physical radiance fidelity.

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